Gene-R1: Reasoning with Data-Augmented Lightweight LLMs for Gene Set Analysis

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Introduction

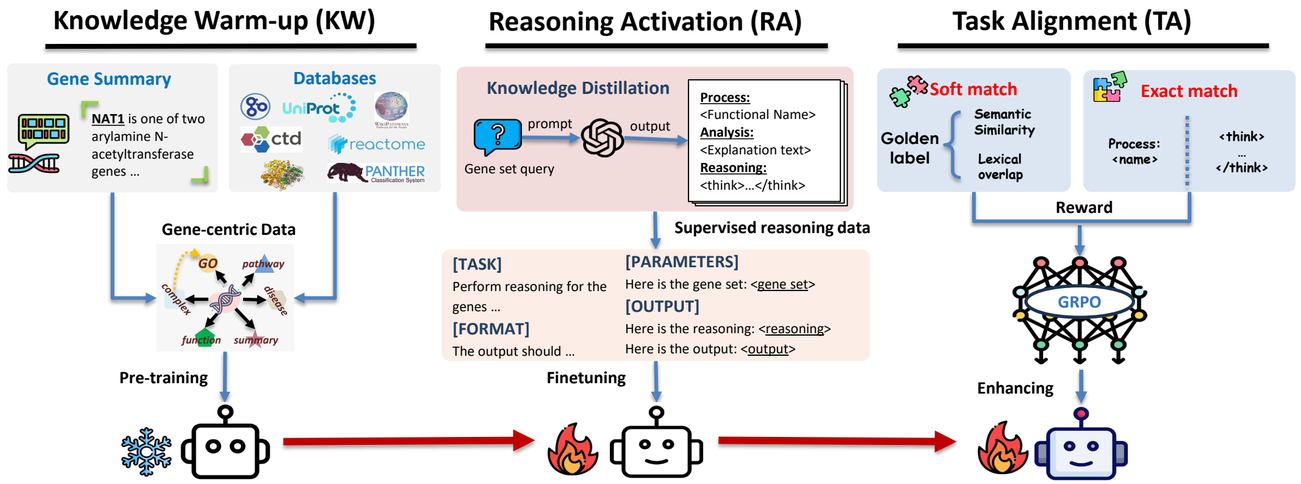
Background

- **Gene-set analysis (GSA)** is a foundational research for revealing the molecular functions associated with groups of genes.
- **Large language models (LLMs)** that incorporate reasoning processes (e.g., GPT-o1, GPT-o3) have shown superior performance across various tasks.
- **Reinforcement learning (RL)** policies have been utilized for fine-tuning open-source LLMs to enable the locally deployment of the systems.

Motivation

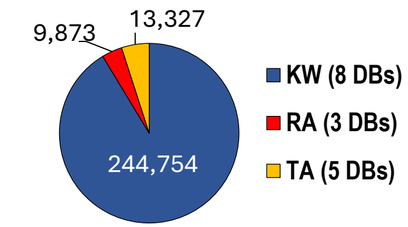
- Most reasoning-based LLMs are **commercial and subscription-based services**.
- **Sensitive data** raises concerns about uploading to platforms where users have limited control over data governance.
- Most fine-tuned LLMs is **trained on general-domain data** and no studies have yet explored reasoning-based solutions **specifically for the GSA task**.

Methodology

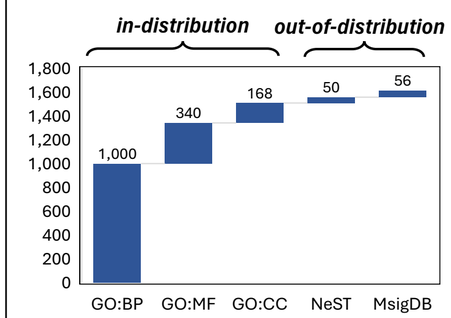


- **KW** equips the model with a foundational understanding of gene symbols and pre-defined biological concept.
  - *Ex. The gene GHRHR is involved in growth hormone signaling and growth regulation.*
- **RA** enables the model to perform reasoning with the step-by-step investigations for gene functions.
  - *Ex. CRY2 gene functions in the regulation of gene expression in response to cellular stress.*
- **TA** equipped with the RL policy enhances both prediction accuracy and output coherence.

Evaluation



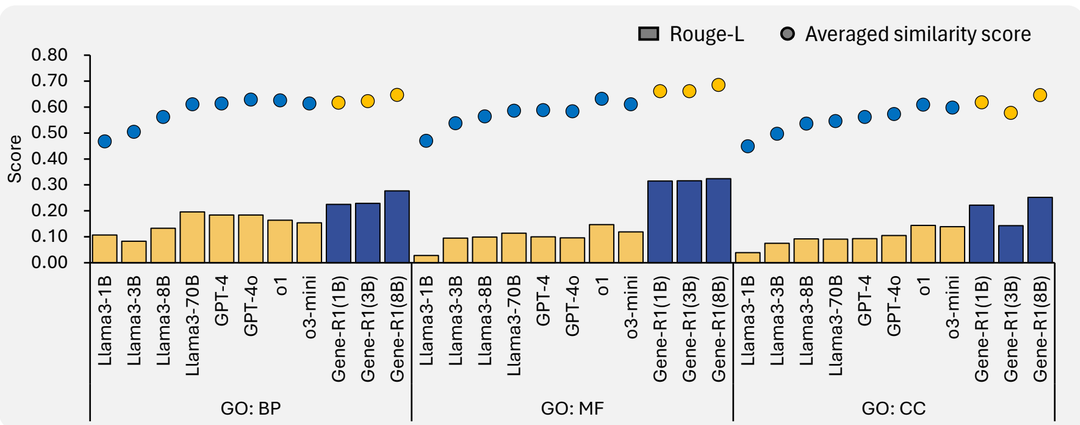
Number of instances used to **fine-tune different modules**



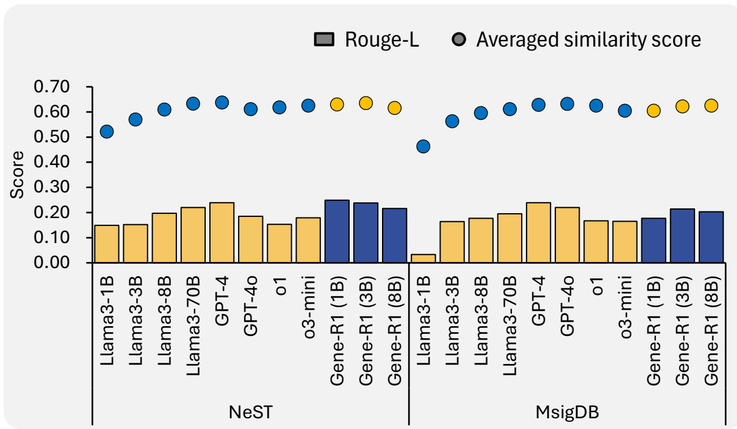
Number of gene sets used to **evaluate the Gene-R1 pipeline**

Experimental Result

In-distribution Results



Out-of-distribution Results



Reference

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